

Numerical Methods - Laboratory Activity 02 Results Summary

Student:James Gabriel C. Estrada | **Dataset:**ReservoirStageLog(96readings)

Parameter	Fitted Value	Standard Error (SE)	t-statistic	P-value
c (Base Level)	14.3775 m	0.0782	183.9	< 1e-15
a (Amplitude)	5.5139 m	0.0812	67.9	< 1e-15
k (Growth Rate)	0.8556 /h	0.0245	34.9	< 1e-15
t0 (Inflection Time)	30.1577 h	0.0512	589.0	< 1e-15

Goodness-of-Fit & Statistical Metrics:

- **Sum of Squared Errors (SSE):** 42.4125 m²
- **Total Sum of Squares (SST):** 2034.0779 m²
- **Coefficient of Determination (R²):** 0.979149
- **Standard Error of Estimate (s):** 0.3864 meters (Degrees of Freedom: 284)
- **Area Under Fitted Curve (scipy.integrate.quad):** 1260.9239 meter-hours (abs error: 4.52e-07)
- **Trapezoid Cross-Check on Raw Data:** 1260.9963 meter-hours

Reading of the Residuals:

The residuals are tightly centered around zero during steady base periods, with minor systematic patterns occurring during the sharp surge phase around July 22. The largest residual (~0.85 m) exceeds the logger's 1 cm resolution, reflecting the limitation of a monotonic four-parameter logistic curve in capturing localized turbulent inflow variations, though the overall high R² (0.979) confirms a robust macroscopic fit.